

DATA ANALYTICS CASE STUDY

Air Quality, Weather & Public Health Impact

Across Indian Cities (2009–2024)

End-to-end analysis: MySQL → Python (Pandas / Seaborn) → Power BI

This project pipelines three independent public datasets — daily weather readings, daily air quality readings, and modeled public-health impact estimates — through SQL, Python, and Power BI to explore how air pollution relates to respiratory and cardiovascular health outcomes across Indian cities.

1. Executive Summary

Objective: quantify the relationship between air pollution (AQI, PM2.5) and public-health burden (respiratory admissions, cardiovascular admissions, mortality) across Indian cities, and identify which cities carry the highest environmental health risk.

Approach: three raw CSV datasets were loaded into a MySQL database, cleaned and joined on Date + City_Code; the same data was independently loaded and merged in Python (Pandas) for exploratory analysis and correlation testing; findings were then built into an interactive four-page Power BI dashboard for stakeholder exploration.

Key Findings

- Air Quality Index (AQI) and the modeled Health Risk Index are strongly correlated ($r = 0.82$) — cities with worse air quality show measurably higher composite health risk.
- PM2.5 is also strongly correlated with Health Risk Index ($r = 0.79$), reinforcing PM2.5 as the pollutant most directly tied to health burden in this dataset.
- Kanpur and Delhi are the most polluted cities in the common-city sample, with average AQI around 345 and 325 respectively — roughly 3x Mumbai's average (~ 120).
- Respiratory and cardiovascular admission estimates are almost perfectly correlated with each other ($r = 0.99$) in this dataset, but only weakly correlated with AQI directly ($r \approx 0.21\text{--}0.24$) — see the Limitations section for why this matters.
- Rainfall shows a weak negative correlation with AQI ($r = -0.13$), consistent with the expected effect of rain clearing particulate matter from the air.

2. Business Problem

Air pollution is a leading environmental health risk in Indian cities, but weather, air-quality, and health-impact data are typically collected and published separately. Without a combined view, it is difficult for public-health planners or researchers to answer practical questions like:

- Which cities have the worst air quality, and how has it moved over time?
- Does higher pollution measurably coincide with higher respiratory/cardiovascular admission estimates?
- Does weather (rainfall, wind) show any relationship with pollution levels?

This project builds the pipeline and dashboard needed to answer those questions from raw data.

3. Data Sources

Dataset	Rows	Columns	Coverage
Air_Quality_updated.csv	33,549	17	26 cities, Jan 2015 – Jul 2020 — PM2.5, PM10, NOx, SO2, CO, O3, AQI, AQI_Bucket
Weather_Report_updated.csv	32,144	11	8 cities, 2009–2020 — temperature, humidity, rainfall, wind, cloud cover, pressure
Health_Report.csv	29,224	16	8 cities, 2015–2024 — modeled respiratory/cardiovascular/asthma/mortality estimates + Health Risk Index

Note: Health_Report figures are dose-response model estimates, not recorded hospital admissions — see Limitations.

4. Methodology

4.1 Data Engineering (MySQL)

All three CSVs were bulk-loaded into MySQL using LOAD DATA LOCAL INFILE for speed (versus row-by-row GUI import). Key steps:

- Created typed tables (air_quality, health_report, weather_report) matching each source schema
- Used NULLIF() during load to convert blank strings to true NULLs in sparse columns (Benzene, Toluene, Xylene)
- Standardized the Date column format across tables (STR_TO_DATE / DATE_FORMAT) so joins on Date would actually match
- Joined all three tables on Date + City_Code and built a reusable master_environmental_report VIEW
- Used window functions (DENSE_RANK) to rank cities by average AQI, and CASE statements to bucket rainfall into No/Light/Heavy Rain categories

4.2 Exploratory Data Analysis (Python)

Independently, all three CSVs were loaded into Pandas, checked for nulls and duplicates, and merged on ['Date','City_Code'] into a single master_df. Seaborn/Matplotlib were used to visualize distributions, city rankings, and cross-variable correlations (see Section 5).

4.3 Dashboard (Power BI)

A star-schema model (Date and City dimension tables related to the three fact tables) powers a 4-page interactive report — Over View, Air Quality, Weather Reports, and Health Report — with 51 visuals total, City/Year/Month slicers, and DAX measures for every KPI card.

5. Data Quality Checks

Check	Air Quality	Weather	Health Report
Duplicate rows	0	0	0
Null values (core columns)	0	0	0
Null values (Benzene/Toluene/Xylene only)	5,623 / 8,041 / 18,109	—	—

The three trace-gas columns (Benzene, Toluene, Xylene) were sparsely recorded at source and were not used in the correlation analysis below for that reason.

6. Exploratory Analysis & Findings

6.1 Which cities are most polluted?

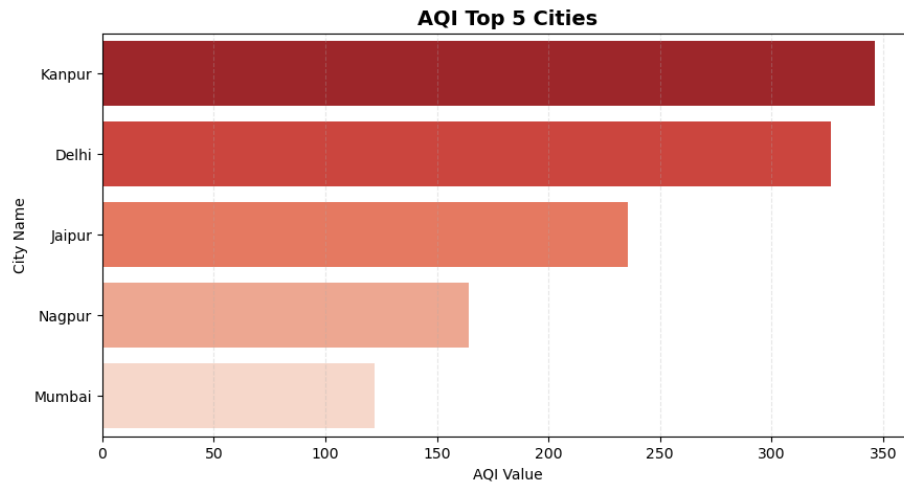


Figure 1 — Top 5 cities by average AQI (common-city sample). Kanpur and Delhi average roughly 2.5–3x Mumbai's AQI.

6.2 How does pollution relate to health risk?

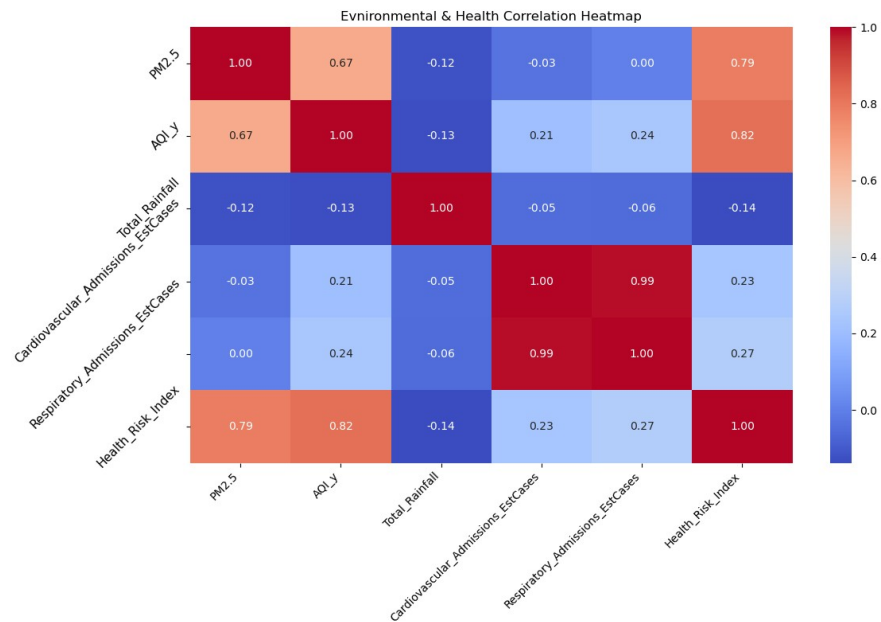


Figure 2 — Correlation matrix across environmental and health variables. AQI ↔ Health Risk Index: $r=0.82$. PM2.5 ↔ Health Risk Index: $r=0.79$.

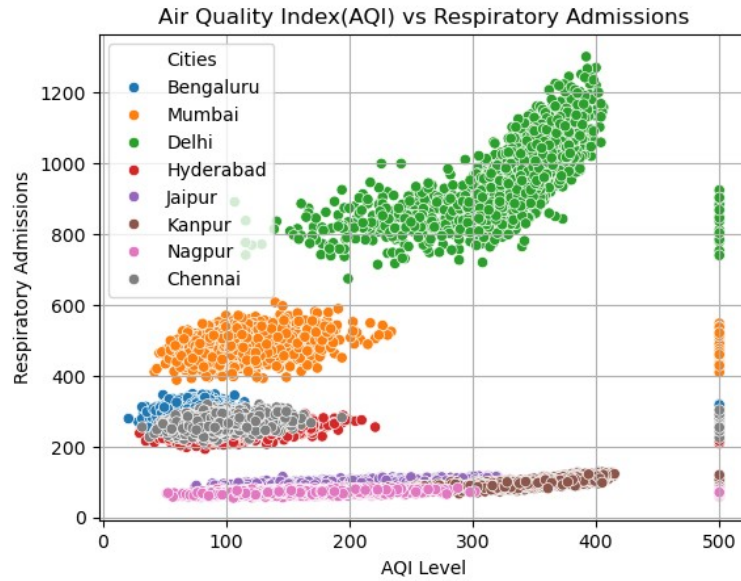


Figure 3 — AQI vs. estimated respiratory admissions, colored by city. The relationship is directionally positive but noisy at the daily level.

6.3 Distribution and spread

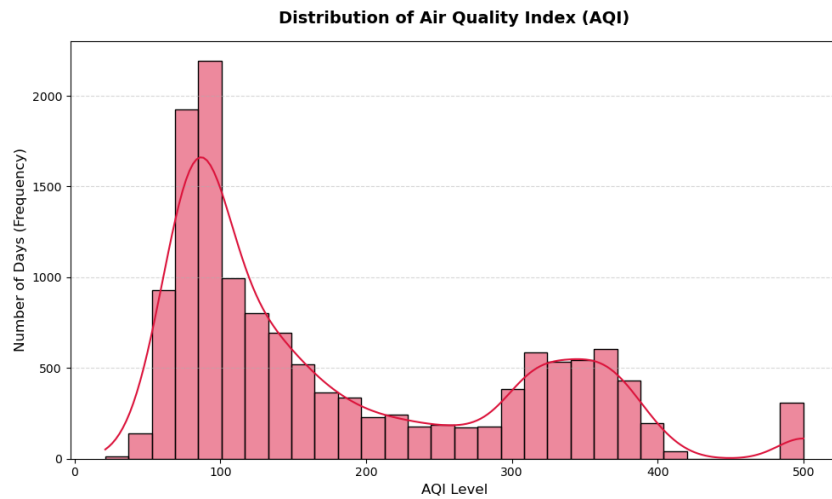


Figure 4 — Distribution of AQI readings across the full dataset — right-skewed, with a long tail of severe-pollution days.

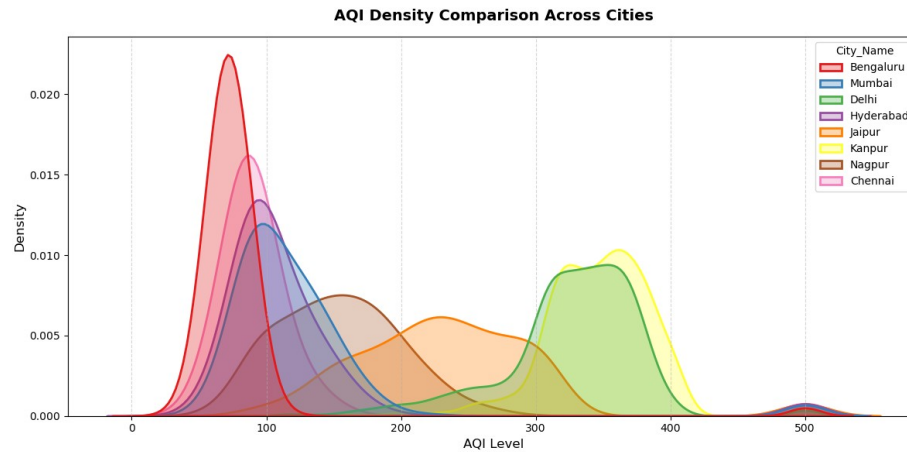


Figure 5 — AQI density comparison across cities — shows how much more spread out and shifted-right the worst cities' pollution distributions are.

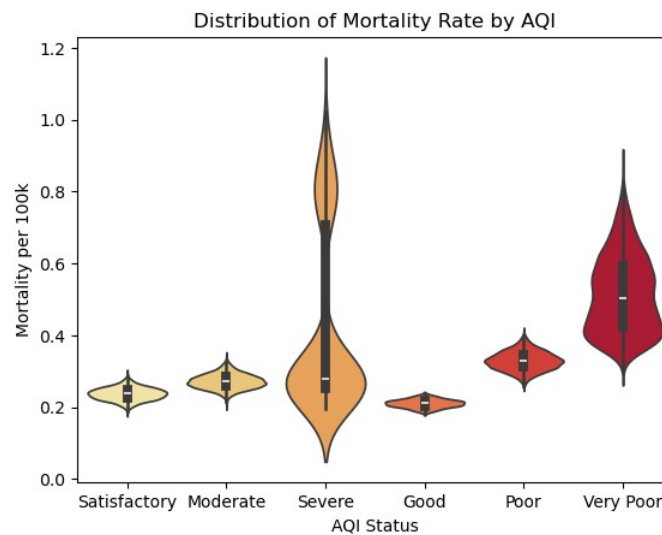


Figure 6 — Distribution of mortality rate (per 100k) by AQI category — higher-severity AQI categories show a wider, higher spread of mortality rate.

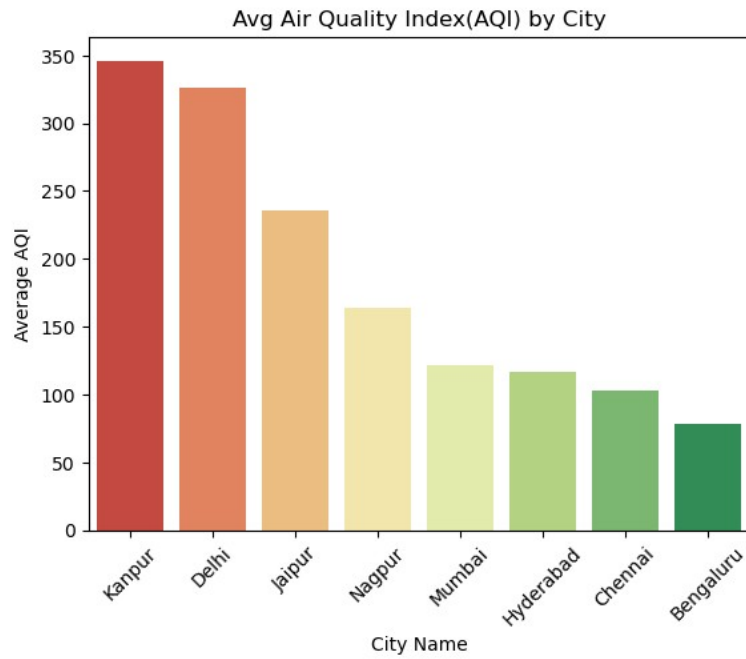


Figure 7 — Average AQI by city, full city set.

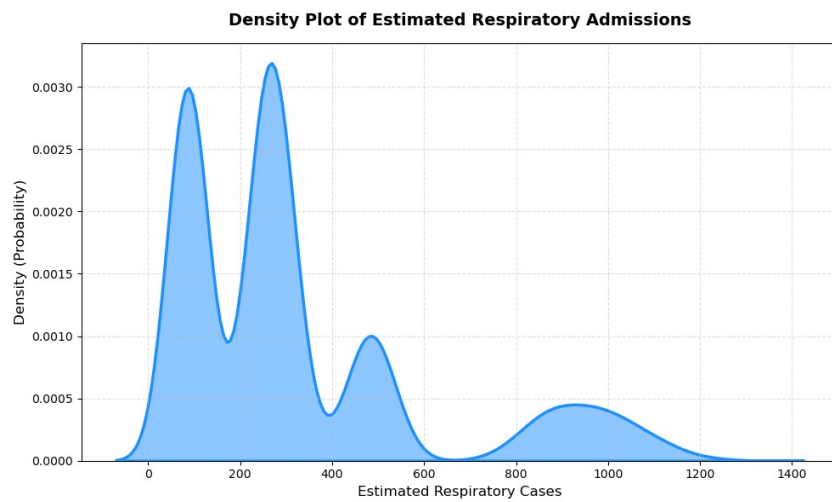


Figure 8 — Density plot of estimated respiratory admissions across all records.

7. SQL Highlights

A selection of queries from the full script (SQL_for_Main_project.sql), showing multi-table joins, window functions, and a reusable view:

City ranking by pollution (window function)

```
SELECT city, ROUND(AVG(aqi),2) AS avg_aqi,  
       DENSE_RANK() OVER (ORDER BY AVG(aqi) DESC) AS pollution_rank  
FROM air_quality  
GROUP BY city;
```

Reusable cross-dataset view

```
CREATE VIEW master_environmental_report AS  
SELECT w.date_raw, w.city_name, w.city_code, w.avg_temp, w.total_rainfall,  
       a.pm2_5, a.aqi,  
       h.aqi_category, h.health_risk_index, h.resp_admissions_est_cases,  
       h.mortality_est_cases  
FROM weather_report w  
JOIN air_quality a ON w.date_raw = a.date_raw AND w.city_code = a.city_code  
JOIN health_report h ON w.date_raw = h.date_raw AND w.city_code = h.city_code;
```

Rainfall categorization (CASE)

```
SELECT city_name,  
       CASE WHEN total_rainfall = 0 THEN 'No Rain'  
            WHEN total_rainfall <= 5 THEN 'Light Rain'  
            ELSE 'Heavy Rain' END AS rainfall_status,  
       COUNT(*) AS total_days  
FROM weather_report  
GROUP BY city_name, rainfall_status;
```

8. Power BI Dashboard

The final deliverable is a 4-page interactive Power BI report (Main_Project.pbix), built on a star schema (Date and City dimension tables related to the three fact tables) with 51 visuals in total.

Page	KPI Cards	Charts	Slicers
Over View	Avg AQI · Avg PM2.5 · Avg Temp · Total Rainfall	Line, Scatter, Treemap, Map	City, Year, Month
Air Quality	Avg AQI · Avg PM2.5 · Avg O3	Bar, Clustered Column, Donut, Line	Year, Month
Weather Reports	Avg Temp · Avg Humidity · Rainfall · Wind	Line, Column, Combo, Clustered Bar	Year, Month

Health Report	Resp. Cases · Mortality · Health Risk Index	Line, Column, 100% Stacked, Area	Year, Month
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[Insert a clean full-page screenshot of each dashboard page here before final submission — export via Power BI Desktop File → Export → PDF, or File → Export → Image for individual pages.]

9. Limitations & Honest Caveats

- Health_Report figures are dose-response model estimates, not recorded hospital admissions data — read as directional risk indicators, not clinical statistics.
- City coverage differs across datasets (7 weather cities, 26 air-quality cities, 8 health cities) — only the cities and date ranges common to all three are valid for cross-dataset comparison.
- Respiratory and cardiovascular admission estimates are correlated at $r=0.99$ with each other, which is unusually high — this most likely reflects both being derived from the same underlying AQI-based formula in the source data rather than two independently measured outcomes, and should be flagged if this dataset is used for anything beyond practice/demonstration.
- Daily-level AQI vs. respiratory-admission correlation is weak ($r \approx 0.21-0.24$) even though city-average AQI vs. Health Risk Index is strong ($r=0.82$) — aggregation level matters, and this gap is worth investigating further rather than glossing over.

10. Skills Demonstrated

Area	Specific Skills
SQL	Schema design, bulk data load (LOAD DATA INFILE), NULL handling, multi-table JOINS, VIEWS, window functions (DENSE_RANK), CASE logic, aggregation
Python	Pandas data loading/cleaning, null & duplicate audits, multi-key merges, Seaborn/Matplotlib (scatter, bar, violin, histogram, KDE, heatmap)
Power BI	Star-schema data modeling, DAX measures, multi-page report design, slicers, 8+ chart types
Analysis	Correlation analysis, city benchmarking, distribution analysis, cross-dataset data quality reasoning